

Appendix: Trigonometric Formulae

- ✗ 1. $\cos^2 A + \sin^2 A = 1$
- ✗ 2. sine formula: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
- ✗ 3. cosine formula: $a^2 = b^2 + c^2 - 2bc \cos A$
- ✗ 4. $\cos (A-B) = \cos A \cos B + \sin A \sin B$
- ✗ 5. $\cos (A+B) = \cos A \cos B - \sin A \sin B$
- ✗ 6. $\cos 2A = \cos^2 A - \sin^2 A$
- ✗ 7. $\sin (A+B) = \sin A \cos B + \cos A \sin B$
8. $\sin (A-B) = \sin A \cos B - \cos A \sin B$
- ✗ 9. $\tan (A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
10. $\tan (A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
11. $\sin 2A = 2 \sin A \cos A$
12. $\sin 2A = \frac{2 \tan A}{1 + \tan^2 A}$
13. $\cos 2A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$
14. $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
15. $\cos^2 A = \frac{1}{2} (1 + \cos 2A)$
16. $\sin^2 A = \frac{1}{2} (1 - \cos 2A)$
17. $2 \cos A \cos B = \cos (A+B) + \cos (A-B)$
18. $2 \sin A \cos B = \sin (A+B) + \sin (A-B)$
19. $2 \sin A \sin B = \cos (A-B) - \cos (A+B)$
20. $2 \cos A \sin B = \sin (A+B) - \sin (A-B)$
21. $\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$
22. $\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$
23. $\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$
24. $\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$

It will be assumed that these formulae are established in the order listed here. In deriving any formula, use may be made of formulae that precede it.

Formal Proofs of Trigonometric Formulae Required for L.C Higher Level

1. $\cos^2 \theta + \sin^2 \theta = 1$

We have already established that any point on the **unit circle** is defined by the coordinates $(\cos \theta, \sin \theta)$.

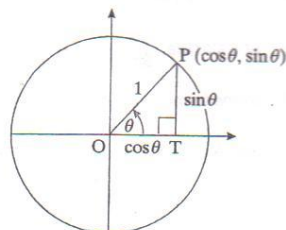
In the given diagram $|OP| = 1$

$$\Rightarrow |OP|^2 = 1$$

$$\Rightarrow \sqrt{(\cos \theta - 0)^2 + (\sin \theta - 0)^2} = 1$$

$$\Rightarrow \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$$

$$\Rightarrow \cos^2 \theta + \sin^2 \theta = 1 \dots (\text{squaring both sides}) *$$



2. Sine Formula:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Proof

Construct a perpendicular h from C to $[AB]$.

$$(i) \frac{h}{b} = \sin A \Rightarrow h = b \sin A$$

$$(ii) \frac{h}{a} = \sin B \Rightarrow h = a \sin B$$

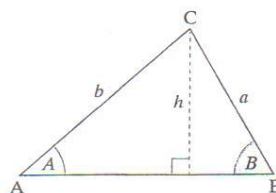
$$\Rightarrow a \sin B = b \sin A$$

Divide both sides by $\sin A \sin B$

$$\Rightarrow \frac{a \sin B}{\sin A \sin B} = \frac{b \sin A}{\sin A \sin B}$$

$$\Rightarrow \frac{a}{\sin A} = \frac{b}{\sin B}$$

Similarly it may be shown that $\frac{b}{\sin B} = \frac{c}{\sin C}$



Sine Rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad \text{or} \quad \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

3. Cosine Formula: $a^2 = b^2 + c^2 - 2bc \cos A$

*Proof

In the $\triangle ABC$, CD is perpendicular to AB.

Let $|CD| = h$ and $|AD| = x$

Thus $|DB| = c - x$

We now apply the Theorem of Pythagoras to the triangles ACD and CDB.

$$\text{In } \triangle ACD: h^2 + x^2 = b^2 \Rightarrow h^2 = b^2 - x^2$$

$$\text{In } \triangle CDB: h^2 + (c - x)^2 = a^2 \Rightarrow h^2 = a^2 - (c - x)^2$$

Equating the two values for h^2 , we get

$$a^2 - (c - x)^2 = b^2 - x^2$$

$$\Rightarrow a^2 - (c^2 - 2cx + x^2) = b^2 - x^2$$

$$\Rightarrow a^2 - c^2 + 2cx - x^2 = b^2 - x^2$$

$$\Rightarrow a^2 = b^2 + c^2 - 2cx$$

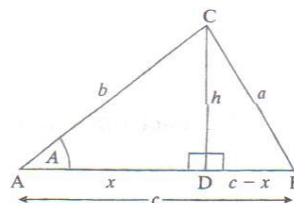
$$\text{But } \frac{x}{b} = \cos A$$

$$\Rightarrow a^2 = b^2 + c^2 - 2c(b \cos A) \Rightarrow x = b \cos A$$

$$\Rightarrow a^2 = b^2 + c^2 - 2bc \cos A$$

Similarly is may be proved that

$$b^2 = c^2 + a^2 - 2ca \cos B \quad \text{and} \quad c^2 = a^2 + b^2 - 2ab \cos C$$



4. $\cos(A - B) = \cos A \cos B + \sin A \sin B$

Let the radii [OP] and [OQ] make angles A and B with the positive x-axis.

$$|\angle POQ| = A - B$$

We now find |PQ| using two different methods:

- the standard formula for the distance between two points
- the cosine rule.

- Using the formula $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, we have:

$$\begin{aligned} |PQ|^2 &= (\cos A - \cos B)^2 + (\sin A - \sin B)^2 \\ &= \cos^2 A - 2 \cos A \cos B + \cos^2 B + \sin^2 A - 2 \sin A \sin B + \sin^2 B \end{aligned}$$

$$\text{But } \sin^2 A + \cos^2 A = 1 \text{ and } \sin^2 B + \cos^2 B = 1.$$

$$\begin{aligned} \Rightarrow |PQ|^2 &= 1 - 2 \cos A \cos B + 1 - 2 \sin A \sin B \\ &= 2 - 2(\cos A \cos B + \sin A \sin B) \dots \textcircled{1} \end{aligned}$$

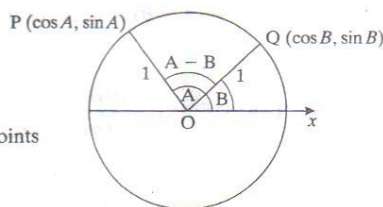
- Now using the cosine rule to find |PQ|, we have:

$$\begin{aligned} |PQ|^2 &= |OP|^2 + |OQ|^2 - 2|OP||OQ|\cos(A - B) \\ &= 1 + 1 - 2(1)(1)\cos(A - B) \dots |OP| = |OQ| = 1 = \text{radius} \\ &= 2 - 2\cos(A - B) \dots \textcircled{2} \end{aligned}$$

Equating the two values for $|PQ|^2$, we have

$$2 - 2(\cos A \cos B + \sin A \sin B) = 2 - 2\cos(A - B)$$

$$\begin{aligned} \Rightarrow -(\cos A \cos B + \sin A \sin B) &= -\cos(A - B) \\ \Rightarrow \cos(A - B) &= \cos A \cos B + \sin A \sin B \dots \textcircled{i} \end{aligned}$$



5. $\cos(A + B) = \cos A \cos B - \sin A \sin B$

To derive the formula for $\cos(A + B)$, we replace B with $(-B)$ in formula (i) on the previous page:

$$\begin{aligned}\cos(A - B) &= \cos A \cos B + \sin A \sin B \\ \Rightarrow \cos[A - (-B)] &= \cos A \cos(-B) + \sin A \sin(-B) \\ &= \cos A \cos B + \sin A (-\sin B) \\ * \Rightarrow \cos(A + B) &= \cos A \cos B - \sin A \sin B \dots\dots (ii)\end{aligned}$$

6. $\cos 2A = \cos^2 A - \sin^2 A$

$$\cos 2A = \cos(A + A) = \cos A \cos A - \sin A \sin A = \cos^2 A - \sin^2 A$$

7. $\sin(A + B) = \sin A \cos B + \cos A \sin B$

$$\cos(A - B) = \cos A \cos B + \sin A \sin B \dots \text{from (i)}$$

To derive the formula for $\sin(A + B)$, we replace A with $(90^\circ - A)$.

$$\begin{aligned}\Rightarrow \cos(A - B) &= \cos[(90^\circ - A) - B] = \cos(90^\circ - A) \cos B + \sin(90^\circ - A) \sin B \\ &= \sin A \cos B + \cos A \sin B \\ \Rightarrow \cos[90^\circ - (A + B)] &= \sin A \cos B + \cos A \sin B \\ * \Rightarrow \sin(A + B) &= \sin A \cos B + \cos A \sin B \dots\dots (iii)\end{aligned}$$

Substituting $(-B)$ for B in formula (iii), we get

$$\begin{aligned}\sin(A - B) &= \sin A \cos(-B) + \cos A \sin(-B) \\ * \Rightarrow \sin(A - B) &= \sin A \cos B - \cos A \sin B \dots\dots (iv)\end{aligned}$$

$$\begin{aligned}\cos(-B) &= \cos B \\ \sin(-B) &= -\sin B \\ \sin(90^\circ - A) &= \cos A \\ \cos(90^\circ - A) &= \sin A\end{aligned}$$

8. $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$$\tan(A + B) = \frac{\sin(A + B)}{\cos(A + B)} = \frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B - \sin A \sin B}$$

We now divide each term in the numerator and denominator by $\cos A \cos B$.

$$\tan(A + B) = \frac{\frac{\sin A}{\cos A} + \frac{\sin B}{\cos B}}{1 - \frac{\sin A}{\cos A} \cdot \frac{\sin B}{\cos B}}$$

$$* \Rightarrow \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \dots\dots (v)$$